BraTS MRI Reconstruction Project

Generated from config `config/formal_train.yaml` on 2026-04-28 15:21:32.

Group Information

- Course: BME AI Project 1
- Group: 第 4 组
- Chinese names: 唐志昊 2022533131
- Note: This report was generated from the final verified experiment outputs and then reviewed before submission.

AI Usage Declaration

AI tools were used only as development assistance for code debugging, experiment orchestration, and draft editing. All implementation decisions, experimental validation, result interpretation, and final submitted materials were manually reviewed and confirmed by the student before submission.

Presentation reminder: AI assistance is explicitly declared in the generated presentation slides, in accordance with the assignment requirement.

Project Objective

The objective of this project is to reconstruct high-fidelity T2-weighted brain MRI slices from AF=5 undersampled k-space using the BraTS dataset. The study is organized into three stages: undersampling simulation, a supervised baseline reconstruction model, and a multi-modal unrolled reconstruction model with explicit data consistency.

Dataset and Preprocessing

- Dataset path: `../bmeaidataset`
- Modalities used: T1, T2
- Slice axis: 2

- Intensity normalization: z-score on non-zero voxels
- Split strategy: patient-level train/validation/test
- Slice selection for this run: 16 central slices per patient
- Data loading strategy: slice-level preloading in RAM to reduce I/O stalls
- Split counts: train=6992, validation=1504, test=1504
- Data split unit: patient-level split to avoid leakage across adjacent slices from the same subject

Methods

Task 1

A 2D random variable-density sampling mask with acceleration factor 5 is generated in k-space. Fully sampled T2 slices are transformed with FFT, masked, and reconstructed with inverse FFT to obtain aliased images. This stage establishes the artifact characteristics that must subsequently be removed by the learning-based reconstruction models.

Task 2

Task 2 uses a U-Net baseline with base channels 24, depth 4, batch size 8, MSE loss, and learning rate 0.0005. `ReduceLROnPlateau` is used for learning rate scheduling.
To improve runtime efficiency on the available RTX 4060 laptop GPU, the implementation uses slice caching, pinned memory, non-blocking GPU transfer, `channels_last`, TF32, and prefetch-friendly dataloading to reduce GPU idle time.

Task 3

Task 3 uses an unrolled U-Net with 2 cascades, data-consistency layers, and multi-modal input consisting of aliased T2 together with fully sampled T1. The loss is `hybrid` with L1 weight 0.7.
The underlying rationale is that T1 provides stable anatomical structure, while the data-consistency layer constrains the network output to remain faithful to the measured undersampled k-space.

Division of Labor

- 唐志昊：完成项目整合、训练实验执行、结果核查、报告整理与提交。

Quantitative Results

- Task 2 improved PSNR from 25.76 dB to 35.18 dB and SSIM from 0.3628 to 0.9271.
- Task 3 improved PSNR from 25.76 dB to 38.12 dB and SSIM from 0.3628 to 0.9660.

- Compared with Task 2, Task 3 changed PSNR by 2.93 dB and SSIM by 0.0389.

Figures and Tables

Dataset Split Chart

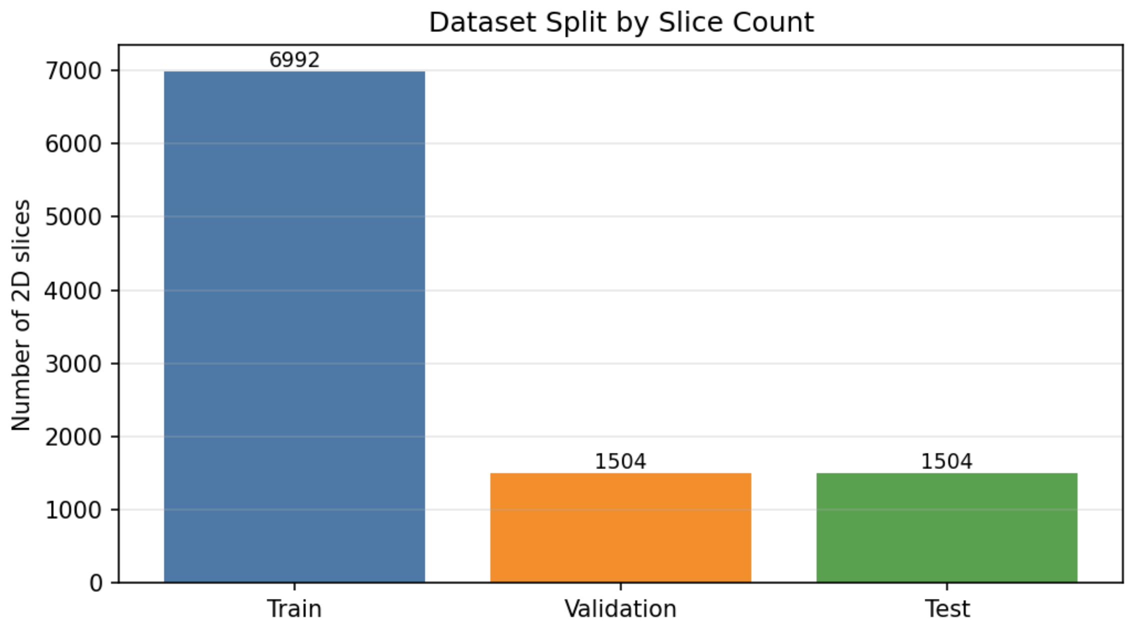

Figure 1: Dataset Split Chart

Metric Comparison Chart

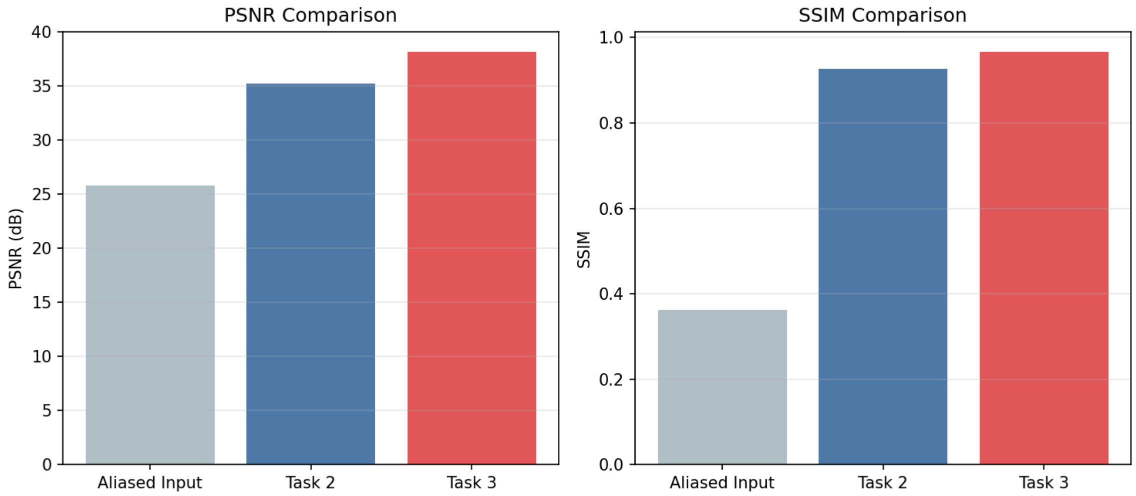

Figure 2: Metric Comparison Chart

Split	Slice Count
Train	6992
Validation	1504
Test	1504

Summary Tables

Dataset Split

Hyperparameters

Setting	Task 2	Task 3
Input	Aliased T2	Aliased T2 + full T1
Backbone	U-Net	Unrolled U-Net + DC
Base channels	24	16
Depth	4	3
Batch size	8	4
Epochs	10	8
Learning rate	0.0005	0.0001
Loss	MSE	hybrid

Quantitative Results

Method	PSNR (dB)	SSIM
Aliased input	25.76	0.3628
Task 2 baseline	35.18	0.9271
Task 3 multi-modal	38.12	0.9660

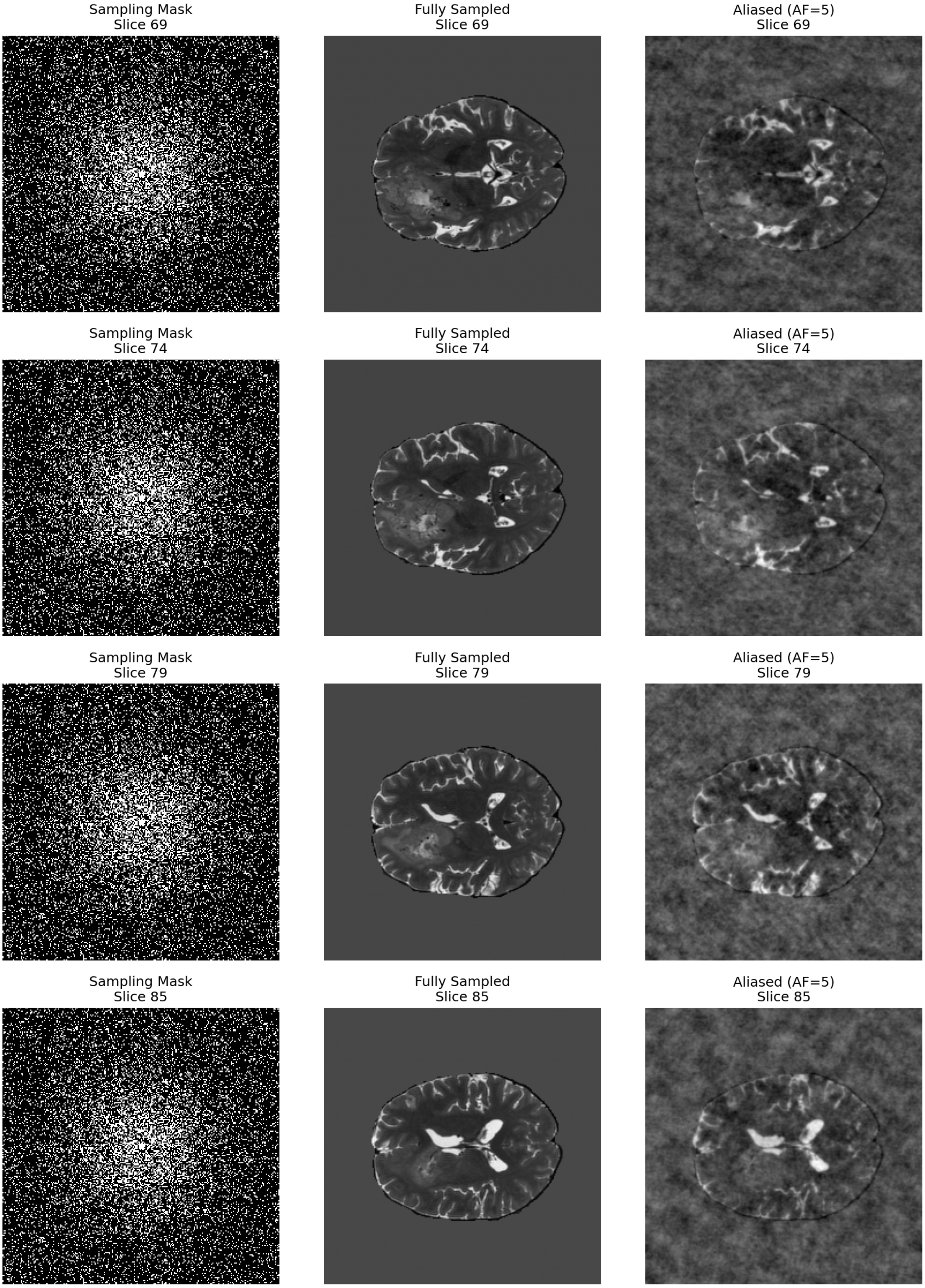

Figure 3: Task 1 Visualization

Method	PSNR Gain (dB)	SSIM Gain
Task 2 baseline	9.43	0.5643
Task 3 multi-modal	12.36	0.6032

Comparison	Value
PSNR gain	2.93 dB
SSIM gain	0.0389

Rank	Patient	Slice	PSNR Before	PSNR After	SSIM Before	SSIM After
1	BraTS-GLI-00077-000	76	23.26	34.69	0.2865	0.9443
2	BraTS-GLI-00077-000	82	23.14	34.74	0.2875	0.9502
3	BraTS-GLI-00088-001	82	24.33	34.82	0.3160	0.9439
4	BraTS-GLI-00077-000	78	23.23	34.82	0.2942	0.9481
5	BraTS-GLI-00077-000	79	23.22	34.84	0.2921	0.9484

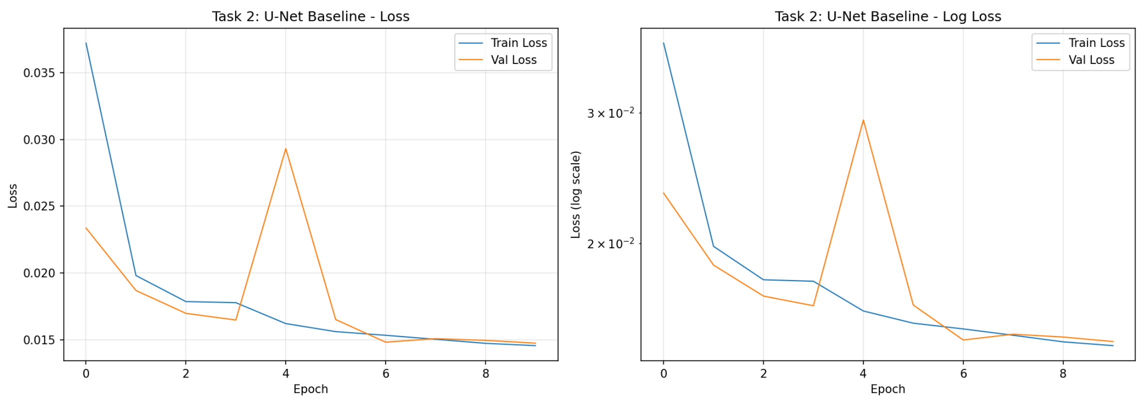

Figure 4: Task 2 Loss Curve

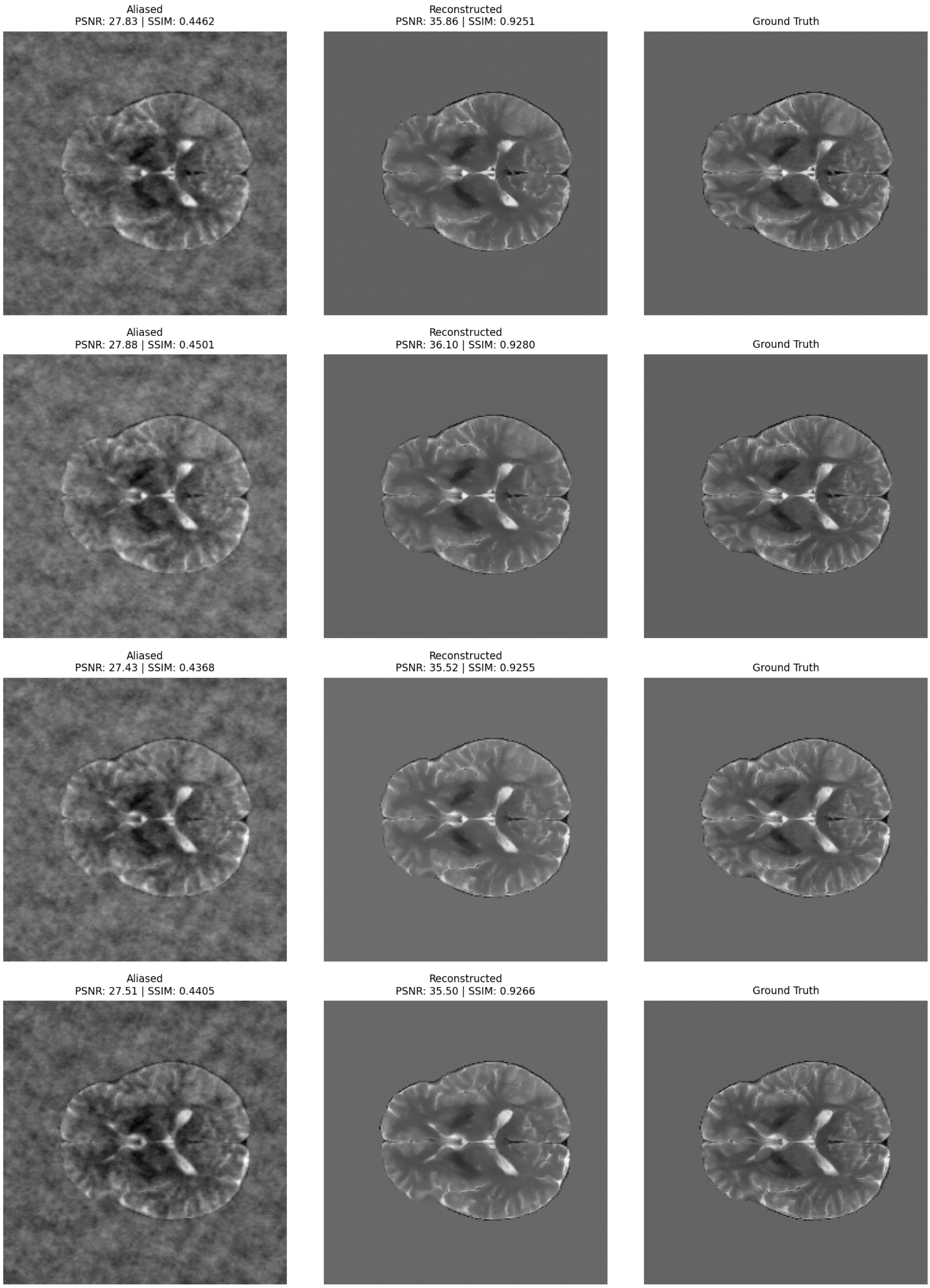

Figure 5: Task 2 Reconstructions

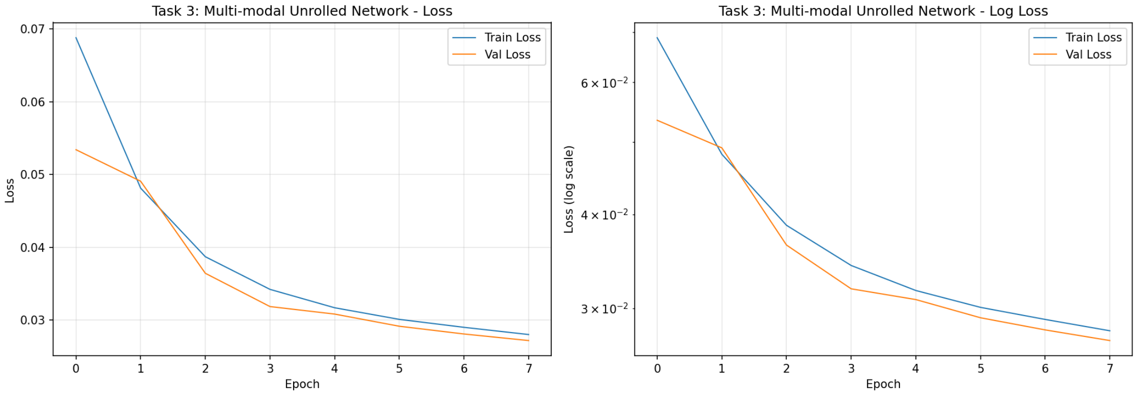

Figure 6: Task 3 Loss Curve

Improvements Over Aliased Input

Task 3 Over Task 2

Worst-case Table

Task 1 Visualization

Task 2 Loss Curve

Task 2 Reconstructions

Task 3 Loss Curve

Task 3 Best Reconstructions

Task 3 Error Analysis

Error Analysis

The 5 lowest-performing Task 3 cases are summarized below. Even in these challenging slices, the reconstructed outputs remain substantially better than the aliased inputs, indicating that the dominant failure mode is relative quality degradation rather than complete reconstruction collapse.

Rank	Patient	Slice	PSNR After	SSIM After
1	BraTS-GLI-00077-000	76	34.69	0.9443
2	BraTS-GLI-00077-000	82	34.74	0.9502
3	BraTS-GLI-00088-001	82	34.82	0.9439
4	BraTS-GLI-00077-000	78	34.82	0.9481
5	BraTS-GLI-00077-000	79	34.84	0.9484

Potential improvements for these cases include stronger edge-preserving loss terms,

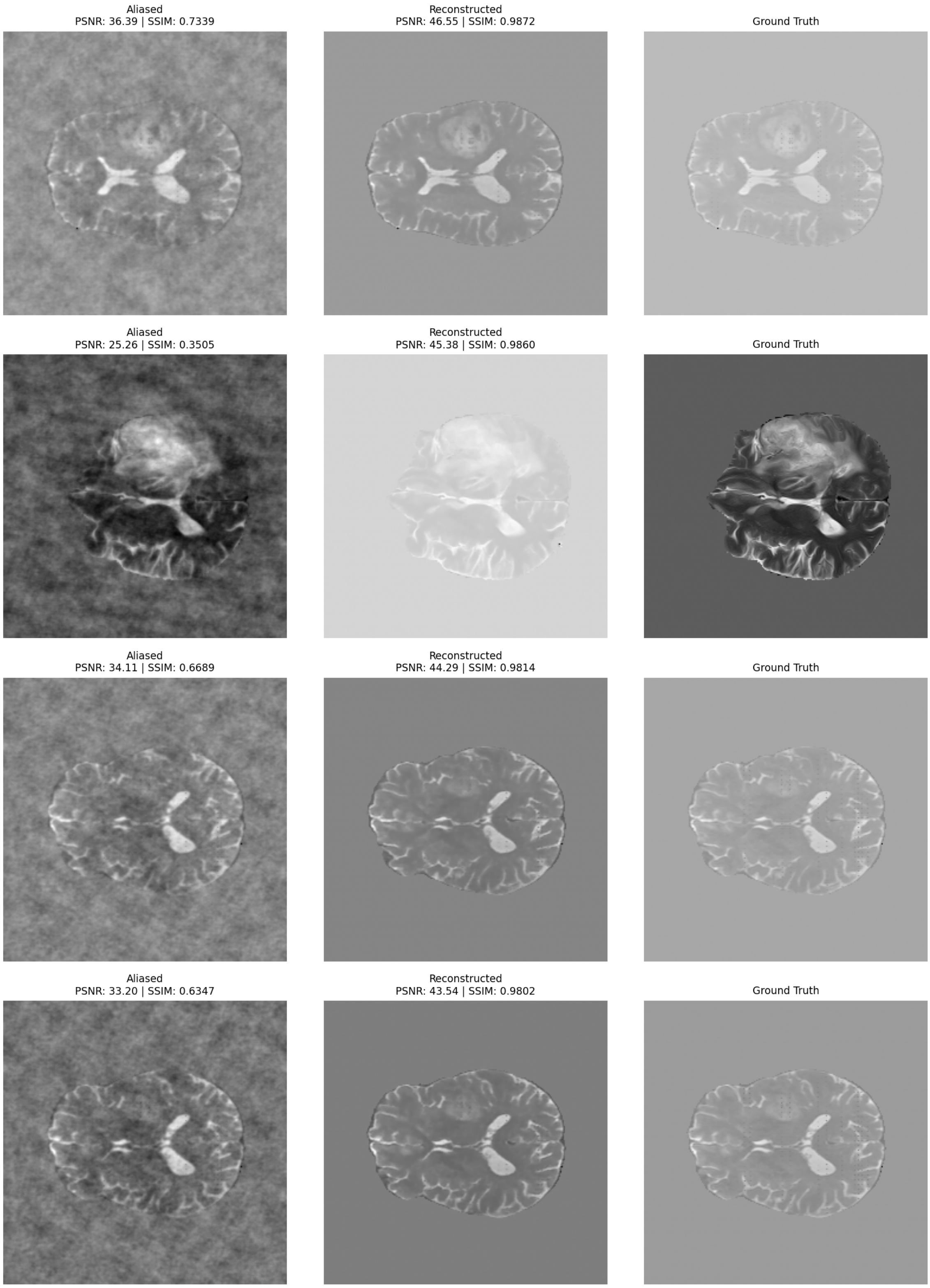

Figure 7: Task 3 Best Reconstructions

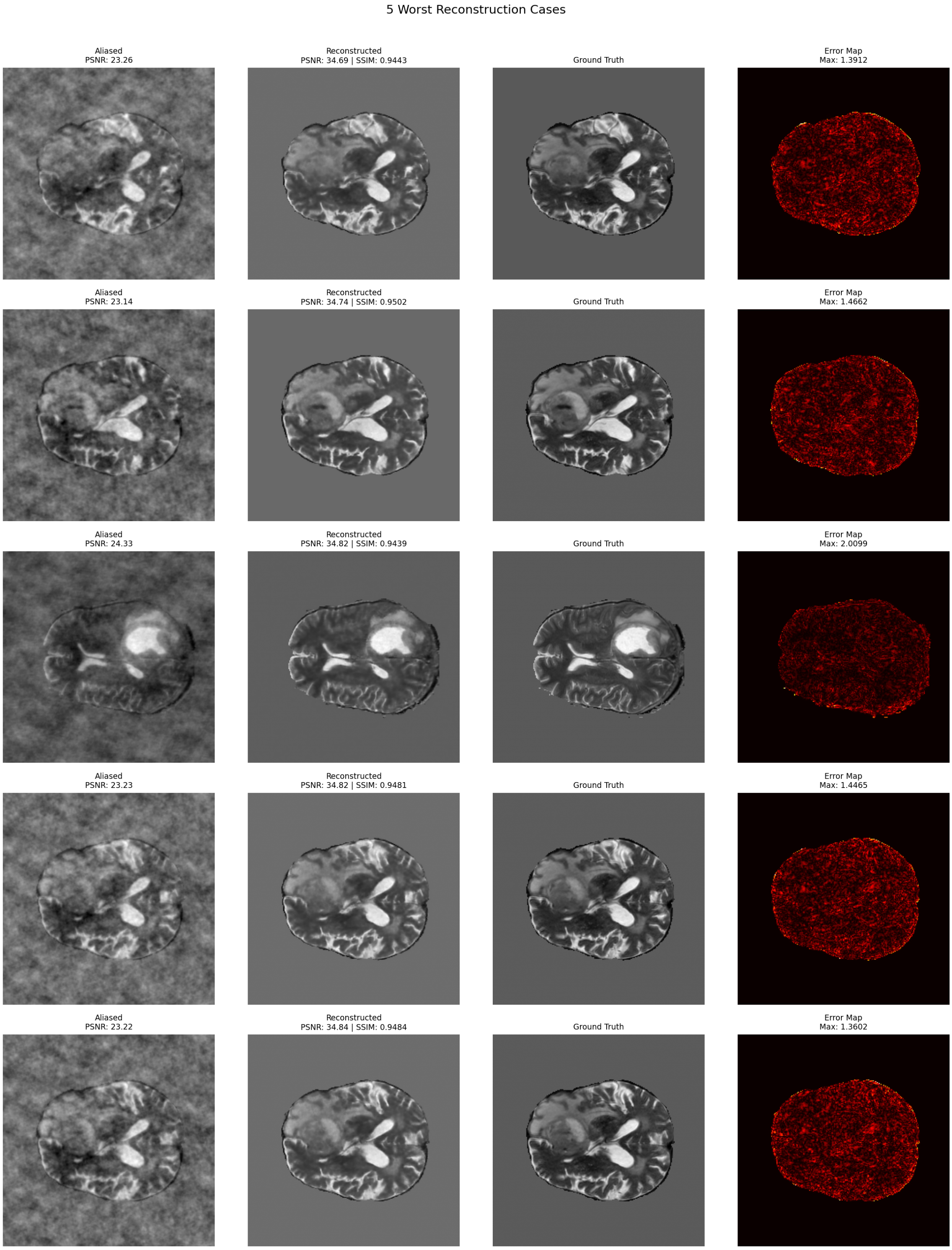

Figure 8: Task 3 Error Analysis

additional cascades if runtime permits, and targeted inspection of slices with complex tumor boundaries.

Execution Status

- Current run status: Success
- Output directory: `outputs_formal`
- Total runtime: 24m 42s
- Last completed stage: assets
- Log file: `outputs_formal/logs/formal_pipeline.log`

Discussion

The final experiment used 16 central slices per patient across all available BraTS patients and produced consistent quantitative improvements over the aliased baseline in both reconstruction settings.

The Task 2 baseline recovered a substantial proportion of the missing image fidelity, improving PSNR by 9.43 dB and SSIM by 0.5643, which confirms that the supervised reconstruction pipeline converged as expected.

Task 3 further improved PSNR by 12.36 dB and SSIM by 0.6032 relative to the aliased input, and outperformed Task 2 by 2.93 dB PSNR and 0.0389 SSIM.

These findings support the central conclusion of the project: incorporating fully sampled T1 structural guidance together with unrolled data-consistency reconstruction yields sharper boundaries, fewer residual artifacts, and stronger quantitative fidelity than a single-modality baseline.

The remaining challenging cases are concentrated in slices with more complex local structure, suggesting that future improvements may be obtained through stronger edge-aware loss functions, a deeper unrolled architecture, or targeted sampling and training strategies for anatomically difficult regions.

Submission Checklist

- Code: prepared
- Report draft: `REPORT.md`
- LaTeX report: `REPORT.tex`
- PDF report: `REPORT.pdf`
- Presentation slides: generated as `slides.tex` and `slides.pdf`